Report 2: UML Class Diagram Report

Classes, Attributes, and Methods

1. Person (Base Class)

- **Attributes:**
  - - personId: String
  - - fullName: String
  - - phoneNumber: String
  - - email: String
  - - address: String
- **Methods:**
  - + updateProfile(): void
  - + getDetails(): String

2. Patient (Inherits from Person)

- **Attributes:**
  - - patientId: String
  - - dateOfBirth: Date
  - - bloodGroup: String
  - - medicalHistory: String
- **Methods:**
  - + registerPatient(): void
  - + bookAppointment(): void
  - + viewMedicalRecords(): List<MedicalRecord>

3. Doctor (Inherits from Person)

- **Attributes:**
  - - doctorId: String
  - - specialization: String
  - - consultationFee: Double
  - - workingHours: String
- **Methods:**
  - + viewSchedule(): List<Appointment>
  - + prescribeMedication(): void
  - + addDiagnosis(): void

4. Receptionist (Inherits from Person)

- **Attributes:**
  - - employeeId: String
  - - shiftTiming: String
- **Methods:**
  - + registerNewPatient(): Patient
  - + scheduleAppointment(): Appointment
  - + generateBill(): Billing

5. Appointment

- **Attributes:**
  - - appointmentId: String
  - - appointmentDate: Date
  - - status: String (Scheduled, Completed, Cancelled)
- **Methods:**
  - + confirmAppointment(): void

- + cancelAppointment(): void
- + reschedule(newDate: Date): void

6. MedicalRecord

- **Attributes:**
  - - recordId: String
  - - visitDate: Date
  - - diagnosis: String
  - - prescription: String
- **Methods:**
  - + createRecord(): void
  - + updateRecord(): void

7. Billing

- **Attributes:**
  - - billId: String
  - - totalAmount: Double
  - - paymentStatus: String (Paid, Pending)
  - - paymentDate: Date
- **Methods:**
  - + calculateTotal(): Double
  - + processPayment(): void

Class 1	Relationship Type	Multiplicity (Class 1 to Class 2)	Class 2	Explanation
Person	Generalization (Inheritance)	1 to 1	**Patient / Doctor / Receptionist**	Patient, Doctor, and Receptionist inherit common attributes and methods from Person.
Patient	Association	1 to 0..*	**Appointment**	A patient can book zero or multiple appointments. Each appointment belongs to exactly 1 patient (1..1).

Class 1	Relationship Type	Multiplicity (Class 1 to Class 2)	Class 2	Explanation
Doctor	Association	1 to 0..*	**Appointment**	A doctor can have zero or multiple appointments scheduled. Each appointment is assigned to 1 doctor (1..1).
Receptionist	Association	1 to 0..*	**Appointment**	A receptionist can schedule multiple appointments.
Patient	Association	1 to 1..*	**MedicalRecord**	A patient has one or more

Class 1	Relationship Type	Multiplicity (Class 1 to Class 2)	Class 2	Explanation
				medical records generated over time. Each record belongs to 1 patient.
Doctor	Association	1 to 0..*	**MedicalRecord**	A doctor creates medical records during consultations.
Patient	Association	1 to 0..*	**Billing**	A patient incurs bills for services rendered. Each bill belongs

Class 1	Relationship Type	Multiplicity (Class 1 to Class 2)	Class 2	Explanation
				to 1 patient.